

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

AN AI-DRIVEN HOTEL RESERVATION AND ROOM MANAGEMENT PLATFORM WITH DYNAMIC PRICING AND DEMAND FORECASTING

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. Identify the Project Vision and Stakeholders

- Define the project's vision, objectives, and identify all key stakeholders involved.

Project Vision

To develop an **AI-Driven Hotel Reservation and Room Management Platform with Dynamic Pricing and Demand Forecasting** that helps hotels automate reservations, optimize room allocation, predict customer demand, and maximize revenue through intelligent pricing strategies while improving the overall guest experience.

Project Objectives

- Develop an online hotel reservation and room management system.
- Implement AI-based dynamic pricing based on demand, occupancy, seasons, and special events.
- Forecast future room demand using historical booking data.

- Automate room allocation and availability management.
- Reduce manual errors and improve operational efficiency.
- Increase hotel revenue through optimized pricing strategies.
- Provide a user-friendly interface for customers and hotel administrators.
- Generate reports and analytics for informed business decisions.

Key Stakeholders

1. **Hotel Owners** – Monitor revenue, occupancy, and overall hotel performance.
2. **Hotel Managers** – Manage room availability, pricing, bookings, and staff operations.
3. **Reception Staff** – Handle reservations, check-in/check-out, and customer requests.
4. **Customers/Guests** – Search rooms, make bookings, and receive pricing recommendations.
5. **System Administrator** – Maintain the platform, manage users, and ensure system security.
6. **Development Team** – Design, develop, test, and maintain the application.
7. **AI/Data Analysts** – Build and improve demand forecasting and dynamic pricing models.
8. **Payment Gateway Providers** – Enable secure online payment processing.
9. **Travel Partners/Booking Platforms** (optional) – Integrate external booking channels to increase reservations.

2. Create Epics and User Stories

- Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.

Requirements Breakdown into Epics and User Stories

Epic 1: User Authentication

- **As a customer**, I want to register an account, **so that** I can access the hotel reservation system.
- **As a customer**, I want to log in securely, **so that** I can manage my bookings.
- **As an administrator**, I want to manage user accounts, **so that** only authorized users can access the system.

Epic 2: Hotel Room Search & Booking

- **As a customer**, I want to search available rooms based on my travel dates, **so that** I can choose a suitable room.
- **As a customer**, I want to book a hotel room online, **so that** I can reserve my accommodation in advance.
- **As a customer**, I want to view my booking history, **so that** I can track my reservations.

Epic 3: Room Management

- **As a hotel manager**, I want to add, update, and delete room details, **so that** room information remains accurate.
- **As a hotel manager**, I want to monitor room availability, **so that** I can efficiently allocate rooms.
- **As a receptionist**, I want to update room status after check-in and check-out, **so that** room availability is always current.

Epic 4: AI-Based Dynamic Pricing

- **As a hotel manager**, I want the system to automatically adjust room prices based on demand, **so that** hotel revenue is maximized.
- **As a customer**, I want to view real-time room prices, **so that** I can make informed booking decisions.
- **As an administrator**, I want to configure pricing policies, **so that** the AI pricing follows business rules.

Epic 5: AI Demand Forecasting

- **As a hotel manager**, I want the system to predict future booking demand, **so that** I can plan room availability effectively.
- **As a hotel owner**, I want occupancy forecasts, **so that** I can improve business planning.
- **As an administrator**, I want demand forecasting reports, **so that** I can analyze booking trends and customer behavior.

Epic 6: Payment Management

- **As a customer**, I want to make secure online payments, **so that** my booking is confirmed instantly.
- **As a customer**, I want to receive a payment receipt, **so that** I have proof of my transaction.
- **As a hotel manager**, I want to verify payment status, **so that** I can manage reservations efficiently.

Epic 7: Reports & Analytics

- **As a hotel owner**, I want to view revenue reports, **so that** I can monitor hotel performance.
- **As a hotel manager**, I want occupancy reports, **so that** I can optimize room utilization.
- **As an administrator**, I want booking analytics dashboards, **so that** I can make data-driven business decisions.

3. Define Acceptance Criteria

- Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

User Story 1: User Registration

As a customer, I want to register an account, so that I can access the hotel reservation system.

Acceptance Criteria:

- **Given** I am a new user, **When** I enter valid registration details and click **Register**, **Then** my account should be created successfully.
- **Given** I enter an email that already exists, **When** I submit the registration form, **Then** the system should display an appropriate error message.
- **Given** mandatory fields are empty, **When** I click **Register**, **Then** the system should prompt me to complete the required fields.

User Story 2: Hotel Room Booking

As a customer, I want to book a hotel room online, so that I can reserve my accommodation in advance.

Acceptance Criteria:

- **Given** rooms are available, **When** I select a room and confirm the booking, **Then** the reservation should be saved successfully.
- **Given** no rooms are available, **When** I attempt to book, **Then** the system should display a "No Rooms Available" message.
- **Given** the booking is successful, **When** the payment is completed, **Then** I should receive a booking confirmation.

User Story 3: Room Management

As a hotel manager, I want to manage room details, so that room information remains accurate.

Acceptance Criteria:

- **Given** I am logged in as a hotel manager, **When** I add a new room, **Then** the room should be displayed in the room list.
- **Given** a room exists, **When** I edit its details, **Then** the updated information should be saved successfully.
- **Given** a room is no longer available, **When** I delete it, **Then** it should be removed from the system.

User Story 4: AI-Based Dynamic Pricing

As a hotel manager, I want the system to automatically adjust room prices based on demand, so that hotel revenue is maximized.

Acceptance Criteria:

- **Given** demand increases, **When** the AI pricing engine runs, **Then** the room price should automatically increase according to pricing rules.
- **Given** demand decreases, **When** the AI pricing engine runs, **Then** the room price should automatically decrease.
- **Given** updated prices are generated, **When** customers search for rooms, **Then** they should see the latest prices.

User Story 5: AI Demand Forecasting

As a hotel manager, I want the system to predict future booking demand, so that I can plan room availability effectively.

Acceptance Criteria:

- **Given** historical booking data is available, **When** the forecasting model is executed, **Then** the system should generate future demand predictions.
- **Given** forecast results are available, **When** I open the dashboard, **Then** I should be able to view demand forecasts.
- **Given** new booking data is added, **When** forecasting is performed again, **Then** the predictions should be updated.

User Story 6: Secure Online Payment

As a customer, I want to make secure online payments, so that my booking is confirmed instantly.

Acceptance Criteria:

- **Given** I have selected a room, **When** I complete a successful payment, **Then** the booking should be confirmed.
- **Given** the payment fails, **When** the transaction is declined, **Then** the booking should not be confirmed and an error message should be displayed.
- **Given** the payment is successful, **When** the transaction is completed, **Then** I should receive a payment receipt.

User Story 7: Reports & Analytics

As a hotel owner, I want to view revenue reports, so that I can monitor hotel performance.

Acceptance Criteria:

- **Given** booking data is available, **When** I open the reports section, **Then** the system should display revenue reports.
- **Given** I select a date range, **When** I generate a report, **Then** the report should show data for the selected period.
- **Given** report generation is complete, **When** I choose to download it, **Then** the report should be exported successfully in PDF or Excel format.

4. Prioritize the Product Backlog

- Organize and prioritize the product backlog based on business value, customer needs, and project dependencies

Priority	Product Backlog Item	Business Value	Customer Need	Dependency
1	User Registration & Login	High	Secure access to the system	None
2	Search Available Rooms	High	Find rooms based on dates and preferences	User Login
3	Hotel Room Booking	High	Reserve rooms online	Search Available Rooms
4	Secure Online Payment	High	Complete bookings safely	Hotel Room Booking
5	Booking Confirmation & Receipt	High	Receive booking confirmation	Online Payment
6	Room Management (Add, Update, Delete Rooms)	High	Manage hotel room inventory	Admin Login
7	Room Availability Management	High	Display real-time room availability	Room Management
8	Customer Booking History	Medium	View past and current bookings	Hotel Room Booking
9	AI-Based Dynamic Pricing	High	Optimize room pricing and maximize revenue	Room Management, Booking Data

10	AI Demand Forecasting	High	Predict future booking demand	Historical Booking Data
11	Reports & Analytics Dashboard	Medium	Analyze occupancy and revenue	Booking, Payment & Forecast Data
12	Booking Cancellation & Modification	Medium	Modify or cancel reservations	Hotel Room Booking
13	Admin/User Management	Medium	Manage user accounts and permissions	User Registration
14	Notification System (Email/SMS Alerts)	Low	Receive booking and payment updates	Booking & Payment Modules

Prioritization Summary

High Priority (Must Have)

- User Registration & Login
- Search Available Rooms
- Hotel Room Booking
- Secure Online Payment
- Booking Confirmation
- Room Management
- Room Availability Management
- AI-Based Dynamic Pricing
- AI Demand Forecasting

Medium Priority (Should Have)

- Customer Booking History
- Reports & Analytics Dashboard

- Booking Cancellation & Modification
- Admin/User Management

Low Priority (Could Have)

- Email/SMS Notification System

This backlog is prioritized based on:

- **Business Value:** Features that directly increase hotel revenue and operational efficiency.
- **Customer Needs:** Core functionalities required for booking and managing reservations.
- **Project Dependencies:** Features are ordered so that foundational modules (login, booking, room management) are completed before advanced AI features and reporting.

5. Plan the Sprint Backlog

- Select high-priority user stories for the first sprint and divide them into actionable development tasks.

Sprint Goal

The goal of Sprint 1 is to develop the core functionalities of the **AI-Driven Hotel Reservation and Room Management Platform**. By the end of this sprint, customers should be able to register, log in, search for available rooms, book a room, and complete a secure online payment. This sprint establishes the foundation for advanced AI features such as dynamic pricing and demand forecasting in future sprints.

User Story 1

As a customer, I want to register an account, so that I can access the hotel reservation system.

Development Tasks

1. Design a user-friendly registration page with input fields for name, email, phone number, and password.
2. Create a database table to store customer information securely.
3. Implement input validation to ensure all mandatory fields are completed.
4. Encrypt passwords before storing them in the database.
5. Display success or error messages based on the registration result.
6. Test the registration process for valid and invalid inputs.

Expected Deliverable

A fully functional registration module that allows new customers to create an account securely.

User Story 2

As a customer, I want to log in securely, so that I can manage my bookings.

Development Tasks

1. Design a secure login page.
2. Implement user authentication using registered credentials.
3. Validate username/email and password.
4. Create user session management after successful login.
5. Display appropriate error messages for invalid credentials.
6. Implement logout functionality.
7. Perform testing for successful and unsuccessful login attempts.

Expected Deliverable

A secure authentication system that allows only authorized users to access the platform.

User Story 3

As a customer, I want to search available rooms, so that I can choose suitable accommodation.

Development Tasks

1. Create a room database containing room details and availability.
2. Design a room search interface.
3. Implement search filters such as check-in date, check-out date, room type, and price.
4. Display only available rooms based on customer preferences.
5. Show room details including price, capacity, and amenities.
6. Test the search functionality using different search criteria.

Expected Deliverable

A room search module that displays available rooms based on customer requirements.

User Story 4

As a customer, I want to book a hotel room online, so that I can reserve my stay.

Development Tasks

1. Design an online booking form.
2. Capture booking details such as customer information, room selection, and stay duration.
3. Verify room availability before confirming the booking.
4. Save booking information in the database.
5. Generate a unique booking ID.
6. Display booking confirmation to the customer.
7. Test booking scenarios to ensure accurate reservation processing.

Expected Deliverable

A booking system that allows customers to reserve available rooms successfully.

User Story 5

As a customer, I want to make secure online payments, so that my booking is confirmed instantly.

Development Tasks

1. Integrate a secure online payment gateway.
2. Validate payment information before processing.
3. Process payments securely.
4. Update booking status after successful payment.
5. Generate a digital payment receipt.
6. Handle payment failures with appropriate error messages.
7. Test successful and failed payment transactions.

Expected Deliverable

A secure payment module that confirms bookings immediately after successful payment.

Sprint 1 Deliverables

At the end of Sprint 1, the following modules will be completed:

- **User Registration Module** – Enables customers to create accounts.
- **User Login Module** – Provides secure authentication and session management.
- **Room Search Module** – Displays available rooms based on selected criteria.
- **Hotel Room Booking Module** – Allows customers to reserve hotel rooms online.
- **Online Payment Module** – Supports secure payment processing and booking confirmation.
- **Booking Confirmation System** – Generates booking IDs and digital receipts.

Definition of Done (DoD)

Sprint 1 will be considered complete when:

- All five user stories are fully implemented.
- Registration and login work without errors.
- Room search displays accurate and available rooms.
- Customers can successfully book hotel rooms.
- Secure online payments are processed correctly.
- Booking confirmation and payment receipts are generated.
- All modules pass functional testing and basic integration testing.
- The application is ready for the next sprint, which will include **AI-based Dynamic Pricing, Demand Forecasting, Room Management, and Reports & Analytics**.

6. Estimate Effort Using Story Points

- Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

Story Point Estimation for Sprint 1

Estimation Technique: Fibonacci Sequence

The **Fibonacci sequence (1, 2, 3, 5, 8, 13, 21)** is used to estimate the effort required for each user story. Story points consider the **complexity of implementation, development effort, uncertainty, and testing effort** rather than the actual time required.

User Story 1: User Registration

User

Story:

As a customer, I want to register an account, so that I can access the hotel reservation system.

Story Points: 3

Justification

- Requires designing a registration page.
- Involves input validation for user details.
- Stores customer information securely in the database.
- Password encryption needs to be implemented.
- Moderate testing is required for valid and invalid registrations.

Since the feature is straightforward with low complexity and minimal dependencies, it is estimated at **3 story points**.

User Story 2: User Login

User

Story:

As a customer, I want to log in securely, so that I can manage my bookings.

Story Points: 5

Justification

- Requires secure authentication.
- Needs password verification and session management.
- Includes handling invalid login attempts.
- Requires logout functionality.
- Security testing is important to ensure authorized access.

Because authentication and session handling increase the complexity compared to registration, this story is estimated at **5 story points**.

User Story 3: Search Available Rooms

User

Story:

As a customer, I want to search available rooms, so that I can choose suitable accommodation.

Story Points: 5

Justification

- Requires designing the room search interface.
- Involves retrieving room details from the database.
- Implements filters such as check-in/check-out dates, room type, and price.
- Displays room availability in real time.
- Requires testing for different search conditions.

This feature has moderate complexity due to database queries and filtering logic, so it is assigned **5 story points**.

User Story 4: Hotel Room Booking

User

Story:

As a customer, I want to book a hotel room online, so that I can reserve my stay.

Story Points: 8

Justification

- Requires booking form development.
- Verifies room availability before confirming the booking.
- Saves reservation details in the database.
- Generates a unique booking ID.
- Sends booking confirmation to the customer.
- Requires integration with room availability management.

Since multiple modules interact and booking accuracy is critical, this user story has high complexity and is estimated at **8 story points**.

User Story 5: Secure Online Payment

User

Story:

As a customer, I want to make secure online payments, so that my booking is confirmed instantly.

Story Points: 8

Justification

- Requires payment gateway integration.
- Processes secure payment transactions.
- Handles payment success and failure scenarios.
- Updates booking status after payment.
- Generates payment receipts.
- Requires extensive security and integration testing.

Because payment processing involves third-party integration, security measures, and comprehensive testing, it is estimated at **8 story points**.

Conclusion

The total effort for **Sprint 1** is estimated at **29 story points** using the **Fibonacci estimation technique**. The sprint focuses on delivering the platform's core functionality—user registration, authentication, room search, booking, and secure payment. Simpler features receive lower story points, while complex features involving integrations, security, and multiple dependencies receive higher estimates. This prioritization helps the development team plan the sprint realistically and deliver a stable foundation for future AI-powered capabilities such as **dynamic pricing** and **demand forecasting**.

7. Present the Sprint Goal and Expected Deliverables

1. Define a clear sprint goal and describe the expected

Sprint Goal and Expected Outcomes

Sprint Goal

The primary goal of **Sprint 1** is to develop the essential features of the **AI-Driven Hotel Reservation and Room Management Platform with Dynamic Pricing and Demand Forecasting**. This sprint focuses on building a secure and user-friendly system that enables customers to register, log in, search for available rooms, make reservations, and complete secure online payments.

The sprint aims to establish a strong foundation for the platform by implementing the core booking workflow before introducing advanced AI functionalities such as dynamic pricing, demand forecasting, room management automation, and business analytics in future sprints.

By the end of Sprint 1, the system should allow users to complete the entire hotel booking process—from creating an account to confirming a reservation—without requiring manual intervention. The delivered features should be reliable, secure, and thoroughly tested to ensure a smooth user experience.

Sprint Goal Objectives

- Develop a secure user registration and authentication system.
- Enable customers to search for available hotel rooms using different search criteria.
- Implement an online room reservation module with real-time availability checking.
- Integrate a secure online payment system for booking confirmation.
- Generate booking confirmations and payment receipts automatically.
- Ensure all implemented features are fully tested and integrated.
- Prepare the platform architecture for AI-based dynamic pricing and demand forecasting in future sprints.

Expected Outcomes

At the end of Sprint 1, the development team is expected to deliver a fully functional prototype that supports the complete hotel reservation workflow. The sprint outcomes are as follows:

1. User Registration Module

Customers will be able to create a new account by entering their personal information. The system will validate user details, securely store customer data, and encrypt passwords to ensure account security.

Outcome:

- Successful account creation.
- Secure storage of user credentials.
- Validation of mandatory registration fields.

2. Secure User Login

Registered customers will be able to log into the system using their credentials. The authentication mechanism will verify user identity and provide secure access to booking services.

Outcome:

- Secure authentication.
- Session management.
- Logout functionality.
- Error handling for invalid login attempts.

3. Room Search Module

Customers will be able to search for available rooms by specifying check-in/check-out dates, room type, and price range. The system will display only rooms that are currently available.

Outcome:

- Real-time room availability.
- Search and filtering options.
- Display of room details and pricing.

4. Online Room Booking

Customers will be able to reserve their preferred rooms through an online booking process. The system will verify room availability before confirming the reservation and generate a unique booking ID.

Outcome:

- Successful room reservation.
- Booking confirmation.
- Automatic booking ID generation.

5. Secure Online Payment

Customers will be able to complete payments using a secure payment gateway. After successful payment, the booking status will be updated automatically and a digital receipt will be generated.

Outcome:

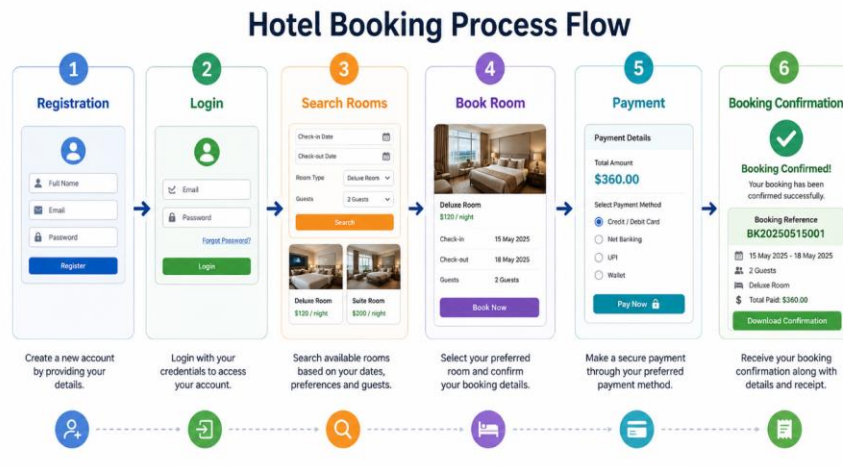
- Secure payment processing.
- Booking confirmation after payment.
- Digital payment receipt generation.
- Error handling for failed transactions.

6. Integrated Booking Workflow

All Sprint 1 modules will work together as one integrated system. Customers should be able to complete the entire booking process without errors.

Outcome:

- Registration → Login → Search Rooms → Book Room → Payment → Booking Confirmation.
- Seamless navigation between modules.
- Consistent data flow across the application.



7. Testing and Quality Assurance

Each developed feature will undergo functional and integration testing to ensure reliability and correctness before deployment.

Outcome:

- Successful execution of test cases.
- Minimal defects.
- Stable application ready for the next sprint.

Sprint Deliverables

The following modules will be completed and demonstrated at the Sprint Review:

- User Registration Module
- User Login and Authentication Module
- Room Search Module
- Hotel Room Booking Module
- Secure Online Payment Module
- Booking Confirmation and Receipt Module
- Integrated Database for Users, Rooms, and Bookings
- Functional Prototype of the Hotel Reservation Platform

Definition of Done (DoD)

Sprint 1 will be considered complete when:

- All selected user stories are fully implemented.
- Registration and login function correctly.
- Room search displays accurate availability.
- Customers can successfully book available rooms.
- Payments are processed securely and booking status is updated.
- Booking confirmations and payment receipts are generated.
- All acceptance criteria are satisfied.
- Functional and integration testing are completed successfully.
- No critical or high-severity defects remain.
- The platform is ready for Sprint 2, where advanced AI features such as **Dynamic Pricing, Demand Forecasting, Room Management Automation, and Analytics Dashboard** will be developed.

Overall Sprint Outcome

At the conclusion of Sprint 1, the **AI-Driven Hotel Reservation and Room Management Platform** will have a complete and functional booking system that enables customers to register, log in, search for rooms, reserve accommodations, and make secure online payments. This successful completion establishes a robust foundation for implementing intelligent AI capabilities and advanced hotel management features in subsequent sprints.

1.sprint deliverables, including the features to be completed at the end of the sprint.

Sprint Deliverables

Sprint deliverables are the functional features and software components that will be completed, tested, and demonstrated at the end of **Sprint 1**. The objective is to deliver a working prototype of the **AI-Driven Hotel Reservation and Room Management Platform** with the core booking functionalities. These deliverables will provide customers with a seamless hotel reservation experience and establish the foundation for advanced AI-based features in future sprints.

1. User Registration Module

A complete registration module will be developed to allow new customers to create an account. The system will collect personal information such as name, email address, phone number, and password. Input validation will ensure that all mandatory fields are completed correctly, while passwords will be securely encrypted before being stored in the database.

Features to be Completed

- Registration form with user-friendly interface.
- Input validation for mandatory fields.
- Duplicate email checking.
- Secure password encryption.

- Customer information stored in the database.
- Registration success and error notifications.

Deliverable

A fully functional customer registration system that allows users to create secure accounts.

2. User Login and Authentication Module

A secure login system will enable registered users to access their accounts. The authentication process will verify user credentials, create secure sessions after successful login, and restrict unauthorized access to the platform.

Features to be Completed

- Login page.
- User authentication.
- Password verification.
- Session management.
- Logout functionality.
- Error handling for invalid login attempts.

Deliverable

A secure authentication system that protects customer accounts and allows authorized access.

3. Room Search Module

Customers will be able to search for available hotel rooms based on their travel requirements. The search feature will retrieve room details from the database and display only rooms that match the selected criteria.

Features to be Completed

- Room search interface.
- Search by check-in and check-out dates.

- Filter by room type and price.
- Display room availability.
- Show room details including amenities and pricing.

Deliverable

A room search system that provides customers with real-time room availability.

4. Hotel Room Booking Module

The booking module will allow customers to reserve available rooms through an online booking process. The system will verify room availability before confirming the reservation and generate a unique booking reference.

Features to be Completed

- Online booking form.
- Room availability verification.
- Booking information stored in the database.
- Automatic booking ID generation.
- Booking confirmation page.

Deliverable

A fully operational hotel room reservation system.

5. Secure Online Payment Module

The payment module will provide customers with a secure method for completing hotel reservations. After successful payment, the booking will be confirmed automatically and the payment details will be recorded.

Features to be Completed

- Payment gateway integration.
- Secure payment processing.
- Payment verification.

- Booking confirmation after payment.
- Digital payment receipt generation.
- Error handling for failed transactions.

Deliverable

A secure online payment system integrated with the booking module.

6. Booking Confirmation System

After a successful booking and payment, the system will generate a booking confirmation containing reservation details. Customers will receive a booking reference number that can be used for future communication.

Features to be Completed

- Automatic booking confirmation.
- Unique booking reference number.
- Display reservation details.
- Payment receipt generation.

Deliverable

An automated booking confirmation and receipt generation system.

7. Database Integration

All customer, room, booking, and payment information will be stored in a centralized database. Proper relationships between tables will ensure accurate and efficient data management

Features to be Completed

- Customer database.
- Room database.
- Booking database.
- Payment database.
- Database relationships.
- Data validation and integrity checks.

Deliverable

A fully integrated database supporting all Sprint 1 functionalities.

8. System Testing and Quality Assurance

Before the sprint concludes, all developed modules will undergo functional, integration, and user acceptance testing to ensure reliability and performance.

Features to be Completed

- Functional testing.
- Integration testing.
- Validation of user stories.
- Bug fixing.
- Performance verification.

Deliverable

A stable, reliable, and tested hotel reservation platform ready for demonstration.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
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Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25